

Hunmin Yang

Personal Website

Senior AI Researcher @ ADD

PhD Candidate @ KAIST

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RESEARCH INTEREST

- **Trustworthy Machine Learning**

- I am broadly interested in the security and safety of machine learning, with a recent focus on multi-modal large language models (LLMs). My previous work includes leveraging generative models for transferable adversarial attacks and building robust AI systems that can operate reliably in diverse environments. I hope to continue making a positive impact on society by enhancing the safety and trustworthiness of AI.

EXPERIENCE

- **Agency for Defense Development (ADD)** Daejeon, Korea
Senior AI Researcher *Jan 2020 - Present*
 - **D-CAM**: Adversarial attack & defense techniques for robust AI
 - **D-GEN**: Synthetic data generation framework for training AI
- **Agency for Defense Development (ADD)** Daejeon, Korea
AI Researcher *May 2017 - Dec 2019*
 - **D-NET**: Large-scale AI inference platform with Hadoop-Spark
- **Agency for Defense Development (ADD)** Daejeon, Korea
Specialized Research Staff (Military Service) *Feb 2014 - May 2017*

EDUCATION

- **Korea Advance Institute of Science and Technology (KAIST)** Daejeon, Korea
PhD in Mechanical Engineering *Sep 2021 - Present*
 - Research Area: Adversarial Machine Learning
 - Advisor: Kuk-Jin Yoon
- **Korea Advance Institute of Science and Technology (KAIST)** Daejeon, Korea
MS in Mechanical Engineering *Feb 2012 - Feb 2014*
 - Research Area: 3D Sound Perception
 - Advisor: Youngjin Park
- **Royal Melbourne Institute of Technology (RMIT)** Melbourne, Australia
Exchange Student (High Distinction) *Feb 2011 - Aug 2011*
- **Korea Advance Institute of Science and Technology (KAIST)** Daejeon, Korea
BS in Mechanical Engineering (Magna Cum Laude) *Feb 2007 - Feb 2012*

PUBLICATIONS

- Se-Yoon Oh, **Hunmin Yang**. Physical Adversarial Attacks and Defenses on Vehicle Detection Models. In *International Conference on Control, Automation and Systems (ICCAS)*, 2025.
- Jongoh Jeong, **Hunmin Yang**, Kuk-Jin Yoon. Boosting Adversarial Transferability with a Generative Model Perspective. In *CVPR Workshop on Generative Models for Computer Vision (GMCV)*, 2025.
- Junhyeong Cho, Kim Youwang, **Hunmin Yang**, Tae-Hyun Oh. Robust 3D Shape Reconstruction in Zero-Shot from a Single Image in the Wild. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025.
- **Hunmin Yang**, Jongoh Jeong, Kuk-Jin Yoon. Prompt-Driven Contrastive Learning for Transferable Adversarial Attacks. In *European Conference on Computer Vision (ECCV)*, 2024. (**Oral, top 2.3%**)
- Junhyeong Cho, Kim Youwang, **Hunmin Yang**, Tae-Hyun Oh. Object-Centric Domain Randomization for 3D Shape Reconstruction in the Wild. In *CVPR Workshop on Foundation Models (WFM)*, 2024.
- **Hunmin Yang***, Jongoh Jeong*, Kuk-Jin Yoon. FACL-Attack: Frequency-Aware Contrastive Learning for Transferable Adversarial Attacks. In *Association for the Advancement of Artificial Intelligence (AAAI)*, 2024.

- Junhyeong Cho, Gilhyun Nam, Sungyeon Kim, **Hunmin Yang**, Suha Kwak. PromptStyler: Prompt-driven Style Generation for Source-free Domain Generalization. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023.
- Naufal Suryanto, Yongsu Kim, Harashta Tatimma Larasati, Hyeon Kang, Thi-Thu-Huong Le, Yoonyoung Hong, **Hunmin Yang**, Se-Yoon Oh, Howon Kim. ACTIVE: Towards Highly Transferable 3D Physical Camouflage for Universal and Robust Vehicle Evasion. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023.
- **Hunmin Yang**, Se-Yoon Oh, Junhyeong Cho. Synthetic Image Generation for Deep Neural Networks. In *NVIDIA GPU Technology Conference (GTC)*, 2023. (**Spotlight**)
- Naufal Suryanto, Yongsu Kim, Hyeon Kang, Harashta Tatimma Larasati, Youngyeo Yun, Thi-Thu-Huong Le, **Hunmin Yang**, Se-Yoon Oh, Howon Kim. DTA: Physical Camouflage Attacks using Differentiable Transformation Network. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.
- Jeonghun Kim, Kyungmin Lee, Hyeonkeun Lee, **Hunmin Yang**, Se-Yoon Oh. Camouflaged Adversarial Attack on Object Detector . In *21th International Conference on Control, Automation and Systems (ICCAS)*, 2021.
- **Hunmin Yang**, Se-Yoon Oh, Taewon Kim, Ki-Jung Ryu. D-GEN: A Deep Learning Data Generation Framework For Artificial Intelligence. In *NVIDIA GPU Technology Conference (GTC)*, 2020.
- Kyungmin Lee, **Hunmin Yang**, Se-Yoon Oh. Adversarial Training on Joint Energy Based Model for Robust Classification and Out-of-Distribution Detection. In *International Conference on Control, Automation and Systems (ICCAS)*, 2020.
- Eunhong Kim, Kanghyun Park, **Hunmin Yang**, Se-Yoon Oh. Training Deep Neural Networks with Synthetic Data for Off-Road Vehicle Detection. In *International Conference on Control, Automation and Systems (ICCAS)*, 2020.
- Hyeonkeun Lee, Kyungmin Lee, **Hunmin Yang**, Se-Yoon Oh. Applying FastPhotoStyle to Synthetic Data for Military Vehicle Detection. In *International Conference on Control, Automation and Systems (ICCAS)*, 2020.
- Kanghyun Park, Hyeonkeun Lee, **Hunmin Yang**, Se-Yoon Oh. Improving Instance Segmentation using Synthetic Data with Artificial Distractors. In *International Conference on Control, Automation and Systems (ICCAS)*, 2020.
- **Hunmin Yang**, Se-Yoon Oh, Ki-Jung Ryu. Accelerating Distributed Deep Learning Inference on multi-GPU with Hadoop-Spark. In *NVIDIA GPU Technology Conference (GTC)*, 2019. (**Oral**)
- **Hunmin Yang**, Se-Yoon Oh, Ki-Jung Ryu. Scalable Distributed Deep Learning Inference on Multi-GPU with Hadoop-Spark. In *NVIDIA GPU Technology Conference (GTC)*, 2019.
- Se-Yoon Oh, **Hunmin Yang**, Ki-Jung Ryu. Optimal Distributed Inference on Multi-GPU Processing System. In *NVIDIA GPU Technology Conference (GTC)*, 2019.
- Se-Yoon Oh, **Hunmin Yang**, Ki-Jung Ryu. Optimal Experimental Design Approach for Machine Learning Process. In *International Conference on Control, Automation and Systems (ICCAS)*, 2017.

PATENTS

Machine Learning & Synthetic Image Generation

- **Hunmin Yang**, Se-Yoon Oh. Training data generation method and apparatus for deep learning model. kr 10-2613781, 2023.
- **Hunmin Yang**, Ki-Jung Ryu, Se-Yoon Oh. Apparatus and method for deep learning based on mixing virtual and real data. kr 10-2198088, 2020.
- **Hunmin Yang**, Ki-Jung Ryu, Se-Yoon Oh. Apparatus and method for learning machine learning models based on virtual data. kr 10-2086351, 2020.
- **Hunmin Yang**, Se-Yoon Oh, Seongbaek Jo. Apparatus and method for enhancing learning capability for machine learning. kr 10-2053202, 2019.
- **Hunmin Yang**, Ki-Jung Ryu, Se-Yoon Oh. Method and apparatus of improving self-supervised learning performance utilizing synthesized data. kr 10-2032519, 2019.
- **Hunmin Yang**, Ki-Jung Ryu, Se-Yoon Oh. Method and Apparatus of adding artificial object for improving performance in detecting object. kr 10-1972095, 2019.
- **Hunmin Yang**, Se-Yoon Oh, Seongbaek Jo. Apparatus and method for generating learning image in game engine-based machine learning. kr 10-1947650, 2019.

AI Security & Adversarial Robustness

- Se-Yoon Oh, **Hunmin Yang**, Hyeongkeun Lee, Kyungmin Lee, Jeonghun Kim. Method and Apparatus for optimizing adversarial patch, computer-readable storage medium and computer program. kr 10-2445215, 2022.
- Hyeongkeun Lee, Jeonghun Kim, Kyungmin Lee, **Hunmin Yang**, Se-Yoon Oh. Method and Apparatus for optimizing adversarial patch, computer-readable storage medium and computer program. kr 10-2414146, 2022.
- Jeonghun Kim, Se-Yoon Oh, Hyeongkeun Lee, Kyungmin Lee, **Hunmin Yang**. Apparatus and method for optimizing adversarial patch based on natural pattern for stealthiness against human vision system. kr 10-2380154, 2022.
- Hyeongkeun Lee, **Hunmin Yang**, Jeonghun Kim, Kyungmin Lee, Se-Yoon Oh. Method, apparatus computer-readable storage medium and computer program for determining adversarial patch position. kr 10-2360070, 2022.

Big Data & Database

- **Hunmin Yang**, Ki-Jung Ryu, Se-Yoon Oh. Method and apparatus of building NoSQL database for signal processing. kr 10-2002360, 2019.
- **Hunmin Yang**, Ki-Jung Ryu, Se-Yoon Oh. Method and apparatus of building inverse index DB for high speed searching of moving picture object. kr 10-2014267, 2019.

HONORS AND AWARDS

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| • Defense Science Award
<i>From the Chief Research Director of ADD</i> <ul style="list-style-type: none">◦ Adversarial Attack Countermeasures for Robust AI-based Image Analysis Systems | Aug 2025 |
| • Selected as Oral Presentation (top 2.3%)
<i>In European Conference on Computer Vision (ECCV)</i> <ul style="list-style-type: none">◦ Prompt-Driven Contrastive Learning for Transferable Adversarial Attacks | Oct 2024 |
| • National Grant for Defense Research and Development
<i>From the Chief Director of DAPA</i> <ul style="list-style-type: none">◦ Synthetic Data Generation for Defense AI | Dec 2021 |
| • Defense Science Award
<i>From the Chief Research Director of ADD</i> <ul style="list-style-type: none">◦ Improving Distributed Multi-GPU Computing for Large-scale Video Analytics | Aug 2019 |
| • High Achievement Award
<i>From the Chief Research Director of ADD</i> <ul style="list-style-type: none">◦ Big Data Platform Development and Synthetic Data Generation | Aug 2018 |
| • Excellent Paper Award
<i>From the Korea Society for Noise and Vibration Engineering (KSNVE)</i> <ul style="list-style-type: none">◦ Sweet Spot Analysis of Linear Array System by Geometrical Approach | Mar 2013 |
| • Scholarship for Academic Excellence
<i>From the Korea Human Resource Development Scholarship Association</i> <ul style="list-style-type: none">◦ Outstanding Academic Performance | Jun 2007 |
| • Scholarship for Academic Excellence
<i>From the Korean Government</i> <ul style="list-style-type: none">◦ Tuition free for all semesters in KAIST (BS & MS) | 2007-2014 |

PROFESSIONAL SERVICES

- **Academic Reviewer**
 - IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)
 - IEEE/CVF International Conference on Computer Vision (ICCV)
 - European Conference on Computer Vision (ECCV)
 - Neural Information Processing Systems (NeurIPS)
 - AAAI Conference on Artificial Intelligence (AAAI)
- **Technology Transfer**
 - Synthetic data generation for training AI models → SI Analytics, Jcorp System, JEIOS
 - Physical adversarial camouflage generation for attacking AI models → SmartM2M
 - Big data platform for large-scale AI model inference → XIIIlab